The Youth Science Journal Of Iraq

Issue #9 June 2026



Satvik M. Bhure

The Youth Science Journal Of Iraq

YSJOI ISSUE #9

©2026 Satvik M. Bhure

First Edition – June 30th, 2026

No part of this publication may be copied or reproduced without written permission from the author(s).

Contact for permissions at YSJOIofficial@gmail.com

Table of Contents:

Cover

Copyright

A Short Editorial

Question I: Is there a limit to how big a particle can get?

Question II: Can light bend?

Question III: What are the strangest animals?

Question IV: Why do we have to sleep?

Question V: Are humans designing their own evolution?

Acknowledgements

Conclusion

A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: Is there a limit to how big a particle can get?

– Dilshad Rami, 13

To begin, let's define a molecule as a chemical compound that can exist on its own while maintaining all of its mechanical properties. You can add more and more atoms to make more and more compounds but does that scale infinitely?

Following the definition stated above, we can assume the largest molecules we know of are large pure crystals such as Quartz which can have a mass of up to 88 tonnes. But could it get bigger? If we used Quartz crystals and continued adding molecules to it forever there will be a few things to consider: firstly, a single grain of dust can result in impurity if landed on the crystal. However, if kept under strict sterilization policies, such found in laboratories, it could grow larger until it collapses under its own weight. Secondly, even if a perfect column of crystals was formed, it would be at a maximum height of 1.9 Km before breaking under tension. Impressive, but not nearly infinite. So, if quartz won't work, what will? Here's a cosmic parallel: If we used Neutron Star Matter, which has a tensile force of 10^{22} MPa, compared to the Quartz's measly 10 MPa, then this will go forever! Will it? Yes, while it will go very far and be very heavy (≈ 1.02 x10^{19} Kg), it will go on forever, per-se. Therefore, yes, if we built a laboratory with infinite length and a weightless interior, then it could go on, hypothetically, forever. Realistically that won't happen.

And even if, by some miracle, you could build this enormous laboratory and build the 'infinite particle' within it, there is of course the limit of gravity. At a point of hugeness, the particle becomes too concentrated on itself and must collapse due to gravity. There is also a philosophical question, or at least one of definition, here; at what point does the particle stop being a particle and becomes something else? Generally a particle must be distinct from its environment; so when something grows so large that it can be called infinite, it may just have to be called a cosmic body instead. So while the math lets us visualize immense crystals of matter, unending and infinite, nature likes chaos a lot more. So semantically and philosophically, this question might have a hard limit, but talking pure math? Well, then technically: there is no limit in sight.

Question II: Can light bend?

– *Asal Haider, 13*

Light, it is the very reason you are able to read this, light is responsible for lighting up the world and allowing us to see. Light is part of the electromagnetic spectrum, it is radiation that travels as both waves and particles (known as photons). It bounces of surfaces and gets absorbed and emitted by other surfaces of different properties, and usually travels in a straight line— it usually does. Well, that raises the question, does light bend?

As for light travel, it does actually bend, in mainly two cases: refraction & gravitational lensing.

Now to explain each case, starting with the former. Refraction is how light bends when it moves from one medium to another, a medium is the physical substance which light travels through, light bends like this because different parts of the wave change speed at different times resulting in a change in direction between mediums. This only happens when the light ray hits the boundary between two mediums at an angle, this causes one of two things, the refraction of light away from an imaginary perpendicular line known as the 'normal' or refraction towards the normal line.

Refraction towards the normal is when the light goes into a denser medium than the one it was already travelling in, the first part of the wave passes through the boundary and its speed decreases as it is in a denser material while the rest of the wave is still moving at the original speed, creating a sort of pivot which causes the light to bend towards the perpendicular normal. The opposite happens when the light goes from a denser medium to a less dense medium which bends the light away from the normal.

Now for the latter option, gravitational lensing is a much more complex concept and is how light bends in space around a black hole. It is the bending of light around a massive object in space with an extremely large mass. Since these bodies have such large masses, bodies like galaxies and black holes, have an immense force of gravity, which according to the relativity theory this gravity isn't just a pulling force, it is actively warping the fabric of space-time around it. Therefore, because light travels in space-time, it will follow the warped path made by the gravitational field and it bends. With perfect alignment this creates a circular halo of light around some objects. This phenomenon is very useful to astronomers who use it to map the regions around it and many more things.

Question III: What are the strangest animals?

– Lana Mustafa, 11

We've all heard of the immortal jellyfish, *Turritopsis dohrnii.* Through a process called transdifferentiation, it transforms its cells (cluster by cluster) from a mature adult state to an immature larval state. But that's not all nature has to offer.

Tardigrades. More commonly known as 'water bears,' these interesting specimens can survive the harshest of conditions, that would even prove fatal to elephants, from near dehydration to boiling temperatures to even outer space. In fact, in 2007, scientists launched them into space. Not only were they exposed to the endless vacuum, but they also endured severe radiation that could kill humans. Those tardigrades came back alive. Most of them, at least. In cases of severe droughts, these microscopic animals can curl up and slow their metabolism down to nearly undetectable levels, preserving water for decades. In this state, they lose 95% of its body water and become a dry capsule known as a tun. As a tun, it can not only survive dehydration, but it can tank temperatures ranging from -200 to 100 degrees Celsius. Although even with all that, outside of its tun state, a tardigrade has a short lifespan, from several months to two years, depending on the species.

Moving on to a fan favorite animal, the axolotl. Everyone has seen or heard of these majestic sea creatures, but very few know what amazing qualities they have. Axolotls can regenerate many parts of their bodies, including severed limbs, sections of their heart and brain, and even the spinal cord. Wherever there is a wound in an axolotl's body, a cluster of cells forms, called a blastema. This structure is highly flexible and can adapt to become whatever cells are needed. Despite this, they can still die of normal causes (old age, poisoning, suffocation, etc.) and are critically endangered.

Our final animal is the mimic octopus. This tentacled mollusc does not just camouflage; it seems to transform. It changes its color and behavior to act as a different animal. Its common disguises include the sea snake, the lionfish, and the flatfish. In combination with its ability to metamorphose, like all octopuses, it possesses remarkable intelligence, which allows it to perfectly fool many organisms.

Question IV: Why do we have to sleep?

– Bariq Abdulmajed, 15

Sleep is one of the most important things we do in life. It not only allows our bodies and minds to recover, but it also helps with learning, memory, and emotional well-being.

During sleep, we also experience dreams, which scientists believe may play a role in processing memories and emotions. The average adult needs around 7 to 9 hours of sleep each night, although this varies from person to person. Some people naturally feel well rested after only 6 hours, while others may need 9 hours to function at their best. But what happens when you do not get enough sleep?

Losing even one hour of sleep can reduce your concentration, reaction time, and decision making abilities. If you stay awake for an entire night, you will not only feel extremely tired but may also struggle with tasks that would normally seem easy.

The longest confirmed time a person has stayed awake is about 11 days. This was done under careful observation and should never be attempted. After being awake for a long time, people can experience hallucinations, memory problems, poor coordination, and microsleeps, which are very short periods of sleep that last from a fraction of a second to several seconds. These can happen without the person even realizing it. Can you recover from lost sleep? You cannot perfectly repay every hour of lost sleep by sleeping the same number of extra hours the next night. However, your body can recover from short periods of sleep loss by getting extra sleep over the following days. The quality of your sleep is also very important. Someone who gets 6 hours of deep, uninterrupted sleep may feel more refreshed than someone who gets 8 hours of poor quality sleep.

So sleep is not a luxury or a boring duty – it is a necessity needed to keep you alive. To make sure that you make sure of your time and sleep, it is essential to keep a good sleep schedule – and as boring as it may sound, avoiding screens for at least an hour before sleep, making sure you sleep comfortably, and meeting with a medical professional is necessary to keep your body running at its optimal velocity; after all, health comes first.

Question V: Are humans designing their own evolution?

– *Zain Rozh*

For thousands of years, humans have searched for ways to overcome their natural limits. We created tools to make our hands stronger, machines to increase our physical abilities, and computers to expand our knowledge. However, today we are developing something far more powerful: technology that can influence the way we think and learn. This raises a fascinating question: are humans designing their own evolution? Evolution traditionally happens through natural selection, where organisms gradually change over generations to survive better in their environment. Humans, however, have started changing their environment instead. Unlike other species, we are not only adapting to the world around us; we are actively creating new possibilities.

Artificial intelligence is one of the biggest examples of this change. AI can already analyze huge amounts of information, recognize patterns, and solve problems at incredible speeds. Some people fear that AI will replace humans, but another perspective is that it may become an extension of human intelligence.
The connection between technology and the human brain is becoming even more interesting. Scientists are researching brain-computer interfaces, which could allow the brain to communicate directly with machines. In the future, such technology might help people with medical conditions, improve learning, or even change how humans interact with information. The idea that our own intelligence could be enhanced by technology was once science fiction, but it is slowly becoming a scientific possibility.

However, designing our own evolution also creates important questions. If humans gain the ability to enhance intelligence or change biological abilities, how far should we go? Should every technological improvement be accepted, or are there limits that we should respect?

The future of humanity may not only depend on natural evolution but also on the choices we make. From simple tools to advanced AI, every invention has changed what humans are capable of achieving. Perhaps the next stage of human evolution will not happen naturally over millions of years, but through the creativity, curiosity, and responsibility of humans themselves.

Acknowledgements

I (Satvik M. Bhure) have many people to thank for this incredible piece of work!

We thank all the scientists who have worked thus far to find all about the world so we can learn about it, and everyone who has supported any member of the YSJOI team in furthering our mission.

Special Thanks To:
Zain Rozh

Asal Haider

Lana Mustafa

Bariq Abdulmajed

Dilshad Rami

Thanks also to the growing team of YSJOI,
where we have welcomed many new members,
all of who had a hand in this incredible issue
and all the work that went into it:

Ali Ahmed Kayani

Parwar Phicri

Ahmed Muhammed

Safa Wisam

Asal Ibrahim

Syed Minhal Abbas

Abdullah Haider

Yousif Muhammed

Hasar Salah

Yara Raqib

Reber Rizgar

And we thank you, reader, for supporting our
mission to help the Iraqi youth where there
has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

You can email us at YSJOIofficial@gmail.com

Or DM us at www.instagram.com/ysjoi_official